

AIGR Gene Hypothesis Deep Research —

S. pombe Rqh1 Four-Way Junction

Helicase Activity (GO:0009378)

Target gene: rqh1 (UniProt Q09811), *Schizosaccharomyces pombe* (strain 972 / ATCC 24843), NCBITaxon:284812 **Seed hypothesis:** *S. pombe* Rqh1 directly unwinds four-way Holliday-junction DNA substrates, supporting GO:0009378 four-way junction helicase activity. **Focus type:** free_text / function-assignment

Summary

Executive verdict: Partially / weakly supported — supported by orthology inference, not by direct experimental evidence. The claim that Rqh1 possesses *four-way junction (Holliday-junction) helicase activity* (GO:0009378) rests entirely on phylogenetic propagation from experimentally characterized RecQ orthologues (human BLM, human WRN, and budding-yeast Sgs1), not on any direct biochemical assay of purified Rqh1 protein. The GO annotation of Q09811 for GO:0009378 carries evidence code **IBA (ECO:0000318, GO_REF:0000033, PAINT/GO_Central)**, with the "with/from" field naming human BLM (P54132), human WRN (Q14191), and Arabidopsis RecQ loci — i.e., the assignment is inferred from ancestral reconstruction, not measured in fission yeast.

By contrast, the **only** experimentally supported molecular-function annotation for Rqh1 is **GO:0043138 "3'–5' DNA helicase activity" (IDA:PomBase)**, which is backed by two independent direct biochemical characterizations of purified Rqh1p: Laursen et al. 2003 ([PMID: 12724426](#)) and Ahmad et al. 2002 ([PMID: 12478586](#)). Both studies demonstrated generic 3'→5' single-strand-displacement helicase activity on standard duplex/tailed substrates and **neither tested a four-way-junction or Holliday-junction substrate**. Targeted literature searches did not identify any primary paper demonstrating in-vitro four-way-junction unwinding or branch migration by purified Rqh1. Direct demonstration of Holliday-junction branch migration and double-Holliday-junction (dHJ) dissolution exists only for the orthologues BLM, Sgs1, and WRN.

Supporting the plausibility of the inference, *S. pombe* in-vivo genetics place Rqh1 firmly in pathways that process branched recombination intermediates — most notably a helicase-activity-dependent suppression of mitotic crossovers (Hope et al. 2007, [PMID: 17353272](#)) — and a computed InterPro domain-architecture comparison confirms that Rqh1 conserves the full RecQ core, including the **RQC winged-helix domain (IPR018982)** and **HRDC domain (IPR002121)** that mediate structure-specific junction recognition in BLM/WRN/Sgs1. These are, however, biological-process (BP) phenotype inferences and structural/evolutionary inferences respectively — they raise prior confidence in the orthology call but do not constitute a direct molecular-function measurement. **Curation lead: retain GO:0009378 as an inferred annotation (IBA/ISS); do not upgrade it to an experimental evidence code; keep GO:0043138 as the directly evidenced molecular function.**

Key Findings

Finding 1 — GO:0009378 for Rqh1 is an orthology inference (IBA), not a direct assay

The UniProt/PomBase annotation of Q09811 assigns GO:0009378 "four-way junction helicase activity" with evidence code **IBA:GO_Central** (phylogenetic inference by ancestral-state reconstruction). The only experimentally supported molecular-function annotation for the same protein is GO:0043138 "3'–5' DNA helicase activity" with evidence code IDA:PomBase, derived from Laursen et al. 2003 ([PMID: 12724426](#)), which characterized purified Rqh1 as a 3'→5' helicase and did not test four-way-junction substrates. Targeted PubMed searches combining "Rqh1" with "Holliday junction," "branch migration," and "branched substrate" biochemistry returned no primary paper describing such an assay. The direct four-way-junction / HJ branch-migration and dHJ dissolution activities are demonstrated only for the orthologues human BLM and *S. cerevisiae* Sgs1.

Laursen et al. 2003 ([PMID: 12724426](#)): "We show that Rqh1, the single *Schizosaccharomyces pombe* homologue, is a 3'-to-5' helicase and exists with Top3 in a high-molecular-weight complex."

On the orthologue BLM ([PMID: 16670433](#)): "*BLM also promotes branch migration of Holliday junctions.*" — This is the directly demonstrated orthologue activity that is the basis of the orthology inference to Rqh1.

Interpretation: the four-way-junction MF term is a well-motivated *inference*, but it is not an experimental fact for the fission-yeast protein.

Finding 2 — In-vivo *S. pombe* genetics support a helicase-dependent role for Rqh1 in processing recombination / four-way-junction intermediates

`rqh1Δ` cells show a ~4-fold increase in crossover frequency during mitotic double-strand-break repair, and this crossover suppression is dependent on Rqh1 helicase activity (Hope et al. 2007, [PMID: 17353272](#)). Multiple additional studies describe Rqh1 as resolving/disassembling recombination joint molecules — e.g. [PMID: 39580958](#) ("RecQ helicase Rqh1 resolves recombination intermediates") and [PMID: 31584934](#) ("Rqh1-mediated disassembly of joint-molecules"). These are mutant-phenotype (IMP/IGI) inferences consistent with in-vivo action on branched/Holliday-junction-containing intermediates, but they do not directly demonstrate four-way-junction unwinding by the purified enzyme.

Hope et al. 2007 ([PMID: 17353272](#)): "*This suppression of crossover formation by Rqh1 is dependent on its helicase activity.*"

Interpretation: the genetics tie Rqh1's helicase function to crossover control — a downstream consequence consistent with junction processing — but map onto a **biological-process** conclusion rather than a molecular-function assay.

Finding 3 — Two independent direct assays show only generic 3'→5' helicase; the GO:0009378 IBA is propagated from experimental BLM/WRN annotations

QuickGO annotation provenance for Q09811 shows GO:0009378 = IBA (ECO:0000318), GO_REF:0000033 (PAINT), with/from human BLM (UniProtKB P54132), human WRN (UniProtKB Q14191), Arabidopsis RecQ loci (AT1G31360, AT4G35740), PANTHER family PTN000344873 — i.e., propagated from experimentally characterized RecQ orthologues, **not** from any Rqh1 experiment. The 3'–5' DNA helicase MF (GO:0043138) carries **two** independent IDA (ECO:0000314) references from PomBase: [PMID: 12724426](#) (Laursen 2003) and [PMID: 12478586](#) (Ahmad, Kaplan & Stewart 2002). Both demonstrate generic 3'→5' DNA helicase

activity for purified Rqh1p; neither tested four-way-junction / Holliday-junction substrates. Ahmad 2002 additionally reports that helicase activity of Rqh1p is only partially required for its function.

Ahmad et al. 2002 (PMID: 12478586): *"Here we show that, like other RecQ family members, the Rqh1p protein displays 3' to 5' DNA helicase activity."*

Ahmad et al. 2002 (PMID: 12478586): *"unlike other RecQ family members, the helicase activity of Rqh1p is only partially required for its function."*

Interpretation: the experimental record for Rqh1 stops at generic 3'→5' helicase. The four-way-junction annotation is entirely a carry-over from experimentally validated orthologues, and the "partially required" caveat warns against treating any single helicase sub-activity as the sole core function.

Finding 4 — Rqh1 conserves the RecQ RQC winged-helix module shared with BLM/WRN/Sgs1, structurally supporting the four-way-junction IBA inference

A computed InterPro domain-architecture comparison (via UniProt REST) across Rqh1 (Q09811), human BLM (P54132), human WRN (Q14191), human RECQ1 (P46063), and *S. cerevisiae* Sgs1 (P35187) shows that Rqh1 shares the full RecQ core: RecQ helicase family (IPR004589), P-loop NTPase (IPR027417), Helicase C-like (IPR001650), RecQ Zn-binding (IPR032284), the **RQC winged-helix domain (IPR018982)**, and **HRDC (IPR002121)** — with BLM, WRN, and Sgs1. The RQC winged-helix is the structural module implicated in structure-specific DNA / Holliday-junction recognition and branch migration in RecQ helicases; allosteric-inhibitor structures of BLM provide "a model for branch migration of Holliday junctions by BLM" (PMID: 33647232).

Interpretation: Rqh1 possesses the machinery known to confer HJ recognition/branch-migration in its orthologues. This raises the prior probability that Rqh1 can act on four-way junctions but is structural/evolutionary evidence — it does not substitute for a direct assay.

Mechanistic Model / Interpretation

The distinction that matters for curation is between **direct molecular-function evidence** and **inference** (phylogenetic, phenotypic, or structural). The evidence for Rqh1 stratifies cleanly:

EVIDENCE FOR Rqh1 four-way-junction helicase (G0:0009378)

DIRECT MF ASSAY on purified Rqh1	
• 3'→5' DNA helicase (G0:0043138) — IDA — https://pubmed.ncbi.nlm.nih.gov/12	
• four-way junction unwinding — NONE (never assayed) x	
IN-VIVO GENETICS (BP-level, phenotype)	
• helicase-dependent crossover suppression — IMP — https://pubmed.ncbi.nlm.nih.gov/12	
• resolves/disassembles joint molecules — IGI/IMP — https://pubmed.ncbi.nlm.nih.gov/31584934	
STRUCTURAL / EVOLUTIONARY (conserved machinery)	
• RQC winged-helix (IPR018982) + HRDC (IPR002121) shared w/ BLM/WRN/Sgs1	
ORTHOLOGUE DIRECT ASSAYS (the basis of the IBA)	
• BLM: HJ branch migration — https://pubmed.ncbi.nlm.nih.gov/16	
• BLM-TopoIIIα-BLAP75: dHJ dissolution — https://pubmed.ncbi.nlm.nih.gov/18	
• Sgs1/BLM RecQ HJ branch-migration enzyme — https://pubmed.ncbi.nlm.nih.gov/24	

IBA propagation (G0_Central / PAINT, G0_REF:0000033, ECO:0000318)
with/from: BLM P54132, WRN Q14191, Arabidopsis RecQ loci

Rqh1 (Q09811) G0:0009378 — INFERRED, not measured

The coherent narrative is that Rqh1 is the **single** RecQ helicase of *S. pombe*, orthologous to Sgs1/BLM/WRN, and functions in a complex with Top3 (topoisomerase III) to process aberrant recombination and replication intermediates during S phase and G2. Its orthologues, when purified, catalyze HJ branch migration and, together with Top3α + BLAP75/RMI, dissolve double Holliday junctions to yield non-crossover products. Rqh1's in-vivo phenotypes (crossover suppression that requires its helicase activity; disassembly of joint molecules; maintenance of circular chromosomes by inhibiting crossovers) are exactly what one would predict if Rqh1 acted on four-way junctions in cells. Conservation of the RQC winged-helix and HRDC modules provides the structural substrate for this activity. **What is missing is the one experiment that would convert inference into fact: an in-vitro assay showing purified Rqh1 unwinding or migrating a defined four-way-junction substrate.**

GO decision table

GO term	Aspect	Current evidence for Rqh1	Recommended curation action
GO:0043138 — 3'-5' DNA helicase activity	MF	IDA ×2 (P 12724426 P 12478586)	Retain — directly evidenced core MF
GO:0009378 — four-way junction helicase activity	MF	IBA only (from BLM/WRN/Sgs1)	Retain as inference (IBA/ISS); do NOT upgrade to IDA/EXP
Recombinational repair / crossover control	BP	IMP/IGI (P 17353272 and others)	Retain as BP — well supported in vivo
Rqh1–Top3 complex	CC/ complex	IDA/IPI (P 12724426)	Retain

Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence / Limitations
12724426 (Laursen 2003)	Direct assay	Qualifies	Does purified Rqh1 have four-way-junction helicase activity?	Rqh1 is a 3'→5' helicase in a complex with Top3; four-way-junction substrate not tested	Purified <i>S. pombe</i> Rqh1p, in vitro	High for 3'→5'; establish absence direct HJ evidence
12478586 (Ahmad 2002)	Direct assay	Qualifies	Same, independent replicate	Rqh1p displays 3'→5' helicase; helicase activity only partially required for function	Purified Rqh1p, in vitro	High; see independent IDA; no 1 substrates
17353272 (Hope 2007)	Mutant phenotype (IMP)	Supports (BP-level)	Does Rqh1 helicase activity control crossovers in vivo?	~4-fold crossover increase in <code>rqh1Δ</code> ; suppression is helicase-dependent	<i>S. pombe</i> mitotic DSB repair	High for genetics; indirect MF
39580958	Mutant phenotype	Supports (BP-level)	Does Rqh1 resolve recombination intermediates?	"Rqh1 resolves recombination intermediates"	<i>S. pombe</i>	Medium phenotype level
31584934	Mutant phenotype	Supports (BP-level)	Does Rqh1 disassemble joint molecules?	Histone deposition antagonizes "Rqh1-mediated disassembly of joint-molecules"	<i>S. pombe</i> arrested forks	Medium phenotype level
16670433 (BLM)	Direct assay (orthologue)	Supports inference	Do RecQ orthologues migrate HJs?	"BLM also promotes branch migration of	Purified human BLM	High for BLM; ba IBA

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence / Limitation
				Holliday junctions"		
24392087	Direct assay (orthologue)	Supports inference	dHJ dissolution mechanism	Top3α required for convergent migration step of dHJ dissolution with BLM	Human BLM/Top3α in vitro	High for ortholog
18390547	Direct assay (orthologue)	Supports inference	BTB complex HJ processing	BLAP75/BLM/TopoIIIα branch-migrate HJ and dissolve dHJ	Human proteins in vitro	High for ortholog
24984776	Review/database	Supports inference	dHJ dissolution as conserved Sgs1/BLM+Top3 function	Defines HJ branch-migration/dissolution for Sgs1/BLM+Top3	Review	Orientat only
33647232	Structural (orthologue)	Supports inference	Structural basis of HJ branch migration	BLM–DNA structures give "a model for branch migration of Holliday junctions"	Human BLM crystallography	High for
InterPro/UniProt REST (computed)	Structural/evolutionary	Supports inference	Does Rqh1 conserve HJ-recognition modules?	Rqh1 shares RQC winged-helix (IPR018982) + HRDC (IPR002121) with BLM/WRN/Sgs1	Computed domain architecture	High for conserva not an activity measure
	Database	Qualifies	What is the evidence code	IBA, ECO:0000318,		

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence / limitations
QuickGO (Q09811, computed)			for GO:0009378?	GO_REF:0000033; with/from BLM/WRN/Arabidopsis RecQ	GO annotation provenance	Definitive evidence code

Conflicts and Alternatives

- **Orthologue carry-over.** The strongest alternative interpretation is that GO:0009378 is a database carry-over from BLM/WRN/Sgs1 rather than a fission-yeast fact. QuickGO provenance confirms this directly (IBA, with/from human BLM P54132 and WRN Q14191). Rqh1 is the *sole* RecQ in *S. pombe*, so this is not intra-organism paralog confusion, but it is cross-species propagation.
- **Helicase-independent functions.** Ahmad 2002 (PMID: 12478586) explicitly reports that Rqh1p's helicase activity is "only partially required for its function," and Top3-related genetics (e.g., PMID: 10572171, PMID: 15340008) show that Rqh1/Top3 lethality relationships involve helicase-dependent and independent components. This cautions against over-weighting any single helicase sub-activity as *the* core function.
- **BP vs MF conflation.** The in-vivo phenotypes (crossover suppression, joint-molecule disassembly) are readily explained by junction dissolution together with Top3 (a topoisomerase-dependent dissolution) rather than by autonomous four-way-junction unwinding. In the orthologue system, robust dHJ processing requires the BLM–TopoIIIα–BLAP75 complex, not BLM alone — so even the orthologue "activity" is partly complex-dependent. The four-way-junction MF term for Rqh1 alone may therefore be biologically narrower than what the enzyme actually does in its physiological Top3 complex.

Limitations and Knowledge Gaps

1. **No direct four-way-junction assay for Rqh1.** *Checked:* PubMed for Rqh1 + Holliday junction / branch migration / branched-substrate biochemistry; PomBase/QuickGO evidence codes. *Why it matters:* this is the single experiment separating inference from fact. *Resolution:* purify recombinant Rqh1 ($\pm$ Top3) and assay on a defined synthetic four-way (X-junction) substrate for unwinding and branch migration.

2. **Rqh1-alone vs Rqh1–Top3 complex activity.** *Checked:* complex composition

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P 12724426
 and orthologue dHJ requirements

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P 24392087
)

P 18390547
Why it matters: the physiological unit is the Rqh1–Top3 dissolvasome; the MF term describes Rqh1 alone. *Resolution:* compare junction processing by Rqh1 alone vs Rqh1–Top3 ($\pm$ Rmi1/BLAP75 orthologue) in vitro.

3. **Whether the IBA should be annotated as ISS with a specific "with" gene.** *Checked:* QuickGO provenance. *Why it matters:* curators may prefer an explicit ISS to BLM/Sgs1 over a PAINT IBA. *Resolution:* curator judgment plus a documented sequence/domain alignment (already supported by Finding 4).
 4. **Direct correspondence between in-vivo crossover suppression and four-way-junction unwinding.** *Checked:* Hope 2007 genetics. *Why it matters:* crossover suppression could operate via D-loop dissolution/SDSA rather than HJ migration. *Resolution:* separation-of-function alleles and substrate-specific in-vitro assays.
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Discriminating Tests

1. **In-vitro four-way-junction unwinding/branch-migration assay** on purified recombinant Rqh1 using a defined synthetic X-junction (fluorescently or radiolabeled), $\pm$ ATP, $\pm$ Mg²⁺. A positive result would directly justify GO:0009378 at IDA; a negative would confirm the term is inference-only for Rqh1.

2. **Rqh1 vs Rqh1–Top3(±Rmi1) comparison** on double-Holliday-junction and mobile-D-loop substrates, mirroring the BLM–TopoIIIα–BLAP75 experiments

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P 24392087
P 18390547
),
 to determine whether junction processing is an autonomous Rqh1 activity or a complex property.

3. **Separation-of-function helicase-dead and junction-binding mutants** (e.g., Walker-A K→I, analogous to rad12-K547I ,

P 10572171
)
 assayed both in vitro (substrate specificity) and in vivo (crossover frequency), to map the crossover-suppression phenotype onto a specific molecular activity.

4. **Structure-guided comparison** of the Rqh1 RQC winged-helix against the BLM branch-migration model

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P 33647232
)
 via AlphaFold/homology modeling to predict junction-binding residues, then test them experimentally.

Proposed Follow-up Actions (Curation Leads — require curator verification)

- **Retain GO:0009378 "four-way junction helicase activity" as an inferred annotation (IBA, ECO:0000318, or ISS to BLM/Sgs1).** Do **not** upgrade to an experimental evidence code (IDA/EXP) — no direct Rqh1 assay exists.
- **Retain GO:0043138 "3'–5' DNA helicase activity" (IDA)** as the primary, directly evidenced molecular function, backed by two independent references

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P 12724426
P 12478586
).

- **Candidate references to verify:**

- P 12478586**

— snippet: *"Here we show that, like other RecQ family members, the Rqh1p protein displays 3' to 5' DNA helicase activity."* (confirms MF is 3'→5', not four-way-junction)
- P 12478586**

— snippet: *"unlike other RecQ family members, the helicase activity of Rqh1p is only partially required for its function."* (caveat on core-function weighting)
- P 17353272**

— snippet: *"This suppression of crossover formation by Rqh1 is dependent on its helicase activity."* (BP-level support)
- P 16670433**

— snippet: *"BLM also promotes branch migration of Holliday junctions."* (orthologue basis of IBA)
- **Suggested question for curators:** should the four-way-junction MF be re-expressed as an ISS to a named orthologue (BLM P54132 / Sgs1 P35187) rather than a PAINT IBA, given the strong domain-architecture conservation documented here?
- **Suggested experiment:** direct in-vitro four-way-junction assay on purified Rqh1 (± Top3), which would resolve the inference-vs-fact gap in a single experiment.

Evidence Base (selected literature)

- **PMID: 12724426** — Laursen et al. 2003. Direct assay: Rqh1 is a 3'→5' helicase in a Top3 complex (basis of GO:0043138 IDA). *No HJ substrate tested.*
- **PMID: 12478586** — Ahmad, Kaplan & Stewart 2002. Second independent 3'→5' helicase IDA; notes helicase activity only partially required.
- **PMID: 17353272** — Hope et al. 2007. Helicase-dependent crossover suppression in vivo (BP support).
- **PMID: 39580958** / **PMID: 31584934** — Rqh1 resolves/disassembles recombination joint molecules (BP support).
- **PMID: 16670433** — BLM promotes HJ branch migration (orthologue direct evidence).

- [PMID: 24392087](#), [PMID: 18390547](#), [PMID: 24984776](#) — BLM/Sgs1 + Top3(+BLAP75/Rmi1) dHJ dissolution and HJ processing (orthologue basis of the conserved-function inference).
 - [PMID: 33647232](#) — BLM structures providing a model for HJ branch migration (structural basis).
 - [PMID: 32085395](#), [PMID: 23161011](#) — Reviews of yeast/human RecQ helicase function (orientation only).
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Report prepared for AI Gene Review hypothesis-level curation. All molecular-function claims are separated into direct Rqh1 assays vs orthologue-derived inference. Computed provenance: QuickGO evidence-code lookup for Q09811 and InterPro/UniProt-REST domain-architecture comparison across Rqh1/BLM/WRN/RECQ1/Sgs1.

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